Lab Program 07: Implementation of Unification

```
def parse(s):
    if '(' not in s:
        return s

    f = s[:s.index('(')]
    inside = s[s.index('(')+1:-1]

    args = []
    cur = ""
    depth = 0

    for ch in inside:
        if ch == ',' and depth == 0:
            args.append(parse(cur))
            cur = ""
        else:
            if ch == '(':
                depth += 1
            elif ch == ')':
                depth -= 1
            cur += ch

    args.append(parse(cur))
    return [f] + args
```

```
def to_str(t):
    if isinstance(t, str):
        return t
    return t[0] + "(" + ",".join(to_str(x) for x in t[1:]) + ")"


def is_var(x):
    return isinstance(x, str) and x in ['x','y','z','X','Y','Z']


def unify(a, b, mgu):

    if a == b:
        return True

    if is_var(a):
        mgu[a] = to_str(b)
        return True

    if is_var(b):
        mgu[b] = to_str(a)
        return True
```

```
    if isinstance(a, list) and isinstance(b, list):

        if a[0] != b[0] or len(a) != len(b):
            return False

        for x, y in zip(a[1:], b[1:]):
            if not unify(x, y, mgu):
                return False

        return True

    return False

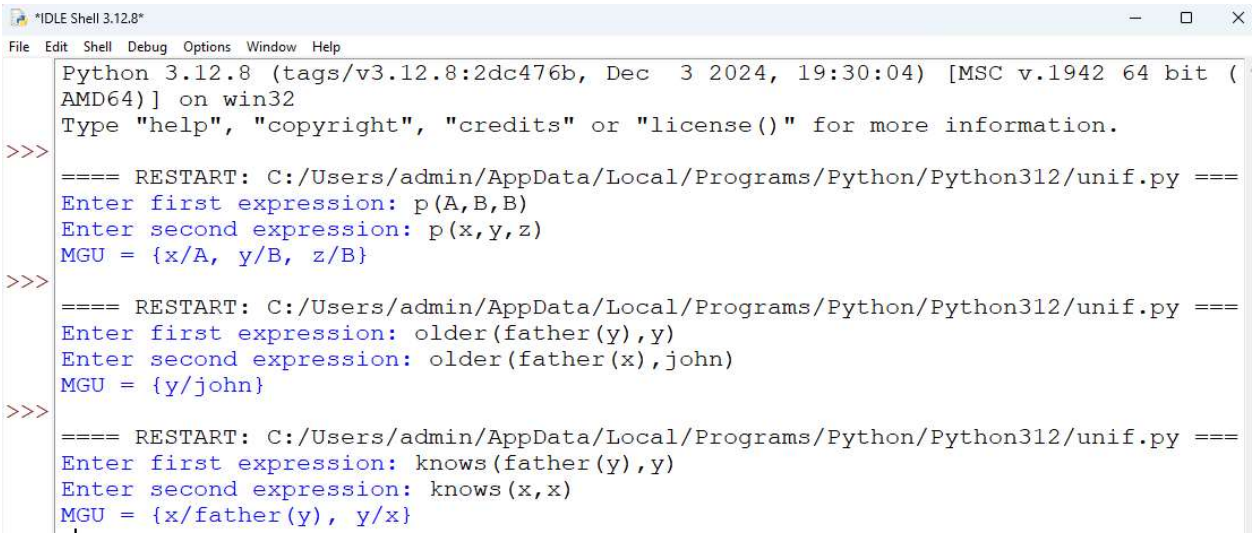


e1 = input("Enter first expression: ")
e2 = input("Enter second expression: ")

mgu = {}

if unify(parse(e1), parse(e2), mgu):
    print("MGU = {", end="")
    print(", ".join(f"{k}/{v}" for k, v in mgu.items()), end="")
    print("}")
else:
    print("Not Unifiable")
```

Output:


IDLE Shell 3.12.8
File Edit Shell Debug Options Window Help
Python 3.12.8 (tags/v3.12.8:2dc476b, Dec 3 2024, 19:30:04) [MSC v.1942 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
==== RESTART: C:/Users/admin/AppData/Local/Programs/Python/Python312/unif.py ===
Enter first expression: p(A,B,B)
Enter second expression: p(x,y,z)
MGU = {x/A, y/B, z/B}
>>>
==== RESTART: C:/Users/admin/AppData/Local/Programs/Python/Python312/unif.py ===
Enter first expression: older(father(y),y)
Enter second expression: older(father(x),john)
MGU = {y/john}
>>>
==== RESTART: C:/Users/admin/AppData/Local/Programs/Python/Python312/unif.py ===
Enter first expression: knows(father(y),y)
Enter second expression: knows(x,x)
MGU = {x/father(y), y/x}